AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Chemistry Paper III

Time: 40 minutes Marks: 10

AKU-EB
Model Paper 2027
for Teaching & Learning only

INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. There are THREE stages in this exam. Each stage contains various tasks. Perform ALL tasks in all stages.
3. Demonstrate the required skills in front of the examiner when instructed to do so.
4. Complete your answers only in the allocated space provided for each task. DO NOT write outside the box. Answers written outside the allocated space will not be marked.
5. Use ONLY black pen or pencil for writing answers. DO NOT use coloured pencils, markers or highlighters.
6. DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers. If you make a mistake, cross it out neatly and write the correct answer beside it.
7. The marks for each task are shown in brackets ( ). Use this information to manage your time efficiently.
8. You may use a simple calculator if you wish.

Practical

Objective: To demonstrate the methods of removing calcium ions from the given hard water samples **A** and **B**.

STAGE I **(Total 1 Mark)**

Task 1

Set the apparatus and perform the experiment. Do not write the method. (Examiner's observation)

_____

_____

STAGE II **(Total 2 Marks)**

Task 2 (1 Mark)

Which substances are responsible for the permanent hardness of water?

_____

_____

Task 3 (1 Mark)

Why is it essential to remove hardness from water?

_____

_____

AKU-EB Model Paper 2027 for Teaching & Learning only

STAGE III **(Total 5 Marks)**

Task 4 (4 Marks)

Write what happens when you perform the tests given in the table below.
Draw an inference from your observations.

Obs. #	Test	Observation	Type of Hardness Removed
1	Sample of hard water **A** + lime water [$Ca(OH)_2$]		
2	Sample of hard water **B** + sodium carbonate solution		

AKU-EB Model Paper 2027 for Teaching & Learning only

Task 5 (1 Mark)

Write a balanced chemical equation for the reaction between sample of hard water **A** and lime water.

_____

_____

School Record **(Total 2 Marks)**

END OF PAPER

Please use this page for rough work

AKU-EB
Model Paper 2027
for Teaching & Learning only